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Department of Computer Science & Engineering

End Term Examination – Nov. 2025

Course: B. Tech.

deepa Sharma

Semester: III

Subject Code: CSE/CSE (AI/CPS/CS/DS)-3005

Subject Name: Signal and Communication Systems

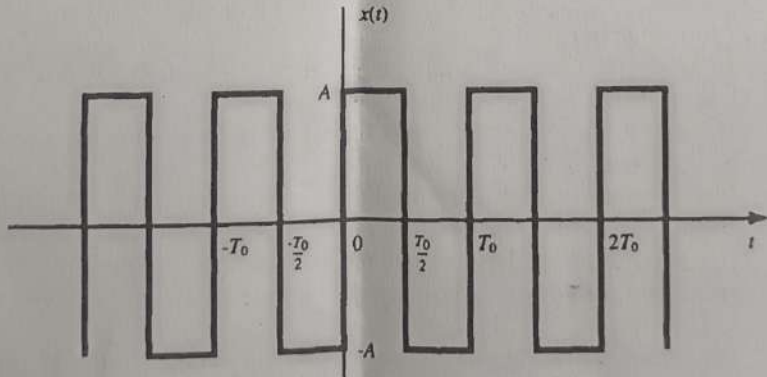
Branch: CSE/ CSE (AI/CPS/CS/DS)

Max. Marks: 60

Time: 03hrs

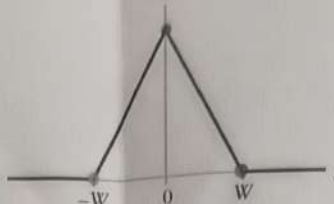
Instructions:

1. All questions are compulsory.
2. Solve all parts of a given question at the same place in sequence.
3. Write correct question number and solve paper neatly and precisely.

Q1.	a. Consider a continuous-time LTI system whose input $x(t)$ and output $y(t)$ are governed by the differential equation: $\frac{d^2 y(t)}{dt^2} - \frac{dy(t)}{dt} - 2y(t) = x(t)$ i. Determine the system transfer function and sketch its pole-zero diagram. ii. Obtain the corresponding impulse response for each of the following cases: (1) The system is stable, (2) The system is causal, (3) The system is neither stable nor causal, (4) The system is both stable and causal. b. Using a well-labeled schematic diagram, explain the generalized communication system and the function of each of its blocks. Additionally, illustrate the various types of communication channels and discuss their common impairments affecting signal transmission.	6
Q2.	a. Consider the periodic square wave $x(t)$ shown in figure.  i. Determine the trigonometric Fourier series of $x(t)$. ii. Also derive complex exponential Fourier series coefficients from the trigonometric Fourier series coefficients. b. Illustrate the Duality Property of the Fourier Transform. Support your explanation with a suitable example demonstrating how the property applies in both time and frequency domains. Also, find the Fourier Transform of the following signal, $x(t) = \sin \omega_0 t$ (using properties) $x(t) = e^{-a t }$ (without using properties)	6

$$\int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt$$

$$A_c \cos(\omega_c t + 2\pi k f \int_0^T m(t) dt)$$

Q3.	a. Which technique of amplitude modulation is regarded as the most optimum for communication systems? Justify your answer by stating relevant reasons. Also discuss the limitations associated with this technique and explain the methods employed to overcome them. b. Consider a message signal $m(t)$ with the spectrum shown in Figure. The message bandwidth $W=1\text{kHz}$. This signal is applied to a product modulator, together with a carrier wave $A_c \cos(2\pi f_c t)$, producing the double sideband suppress carrier (DSB-SC) modulated wave $s(t)$. This modulated wave is next applied to a coherent detector.	6 6
	$M(f)$  -W 0 W Assuming perfect synchronism between the carrier waves in the modulator and detector, determine the spectrum of the detector output when: - the carrier frequency $f_c = 1.25\text{ kHz}$ and - the carrier frequency $f_c = 0.75\text{ kHz}$. What is the lowest carrier frequency for which each component of the modulated wave is uniquely determined by $m(t)$?	6 6
Q4.	a. With the help of a neat block diagram, describe the indirect method of generating a frequency-modulated (FM) signal. A carrier of frequency 20 MHz is frequency-modulated by a sinusoidal message signal, producing a maximum frequency deviation of 100 kHz. If the modulating signal frequency is 1 kHz, calculate the modulation index and transmission bandwidth of the resulting FM signal using Carson's rule. b. Explain the complete process of converting an analog signal into its digital form, highlighting each stage involved in the transformation. Further, illustrate in detail the operation of any one digital modulation technique used for transmitting the resulting digital data over a communication channel.	6 6
Q5.	a. A Compact Disc (CD) records audio signals digitally by using PCM. Let the audio signal bandwidth be 15 kHz. - Determine the sampling rate if the signal is to be sampled at a rate 2/9 above the Nyquist rate. - If the Nyquist samples are uniformly quantized into 65,536 levels and then binary-coded, determine the number of binary digits required to encode a sample. - If the audio signal has average power of 0.1 W and peak voltage of 1 V, find the resulting ratio of signal to quantization noise (SNR) of the uniform quantizer output in part (ii). - Determine the number of binary digits per second (bit/s) required to encode the audio signal. - Find the minimum bandwidth required to transmit the PCM signal. b. Explain the concept of multiple access technique in communication systems? Provide a detailed comparative analysis of the performance of TDMA, FDMA, and CDMA schemes in cellular networks with respect to bandwidth efficiency, interference management, capacity, and system complexity.	6 6